

Legal Documents Query Application for Vietnamese Law Using LLM and RAG Techniques

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Abstract

This paper introduces the development of a Legal Documents Query Application for Vietnamese law, leveraging Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) techniques. It outlines the methodologies employed, addresses key challenges, and highlights the significance of the application in improving the accuracy and efficiency of legal document retrieval in specialized domains. By combining cutting-edge AI techniques with domain-specific optimizations, the proposed solution aims to enhance access to legal information for a wide range of users.

Keywords

Legal Document Retrieval, Large Language Models, Retrieval-Augmented Generation, Vietnamese Law, Natural Language Processing, Legal AI, Semantic Search

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1 Introduction

Large Language Models (LLMs) have demonstrated exceptional capabilities across a variety of natural language processing (NLP) tasks, including text summarization, translation, and question answering. Despite their success in general domains, achieving high precision in specialized fields, such as law and finance, remains a significant challenge. This is particularly evident in languages like Vietnamese, which feature unique grammatical structures and linguistic complexities. This paper explores the application of LLMs in

the legal domain, with a focus on developing a solution for querying Vietnamese legal documents.

The retrieval and querying of legal documents in Vietnamese present several unique challenges. The language's intricate grammar and rich vocabulary complicate the retrieval of accurate and contextually relevant information. Moreover, the specialized terminology and complex legal concepts inherent in legal documents require advanced domain-specific knowledge. These factors make formulating effective queries and obtaining precise, concise answers a demanding task, especially within the context of Vietnamese legal documentation.

To address these challenges, this paper proposes a novel approach that integrates LLMs with Retrieval-Augmented Generation (RAG) techniques, tailored specifically to the nuances of the Vietnamese language and legal domain. This application is designed to provide accurate, efficient, and context-aware access to legal information, benefiting legal professionals, students, and the general public in Vietnam.

1.1 Querying Legal Documents Problems

Querying legal documents presents unique challenges due to the nature of the legal domain, the volume of information available, and the high standards of accuracy and confidentiality required. These challenges impact the efficiency and reliability of legal document retrieval, emphasizing the need for specialized techniques in managing, processing, and querying legal databases.

- (1) *Volume of Legal Document Databases*: Legal databases encompass a vast array of materials, including legislation, case law, contracts, and regulatory guidelines, spanning multiple jurisdictions, historical periods, and legal areas. This sheer volume increases computational demands, necessitates robust indexing, and complicates the retrieval of precise results. Effective systems must balance comprehensive retrieval with minimizing irrelevant information to avoid overwhelming users.
- (2) *Time Constraints*: Legal professionals, including lawyers and researchers, often operate under tight deadlines for tasks such as preparing for court cases, drafting legal documents, or advising clients. Traditional methods of legal research, which often require manual searches through dense texts, are time-intensive and inefficient. Automated querying systems must provide fast, accurate, and contextually relevant

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results while ensuring high-quality information retrieval under strict time constraints.

- (3) *Language Complexity*: Legal language is characterized by specialized terminology, archaic phrasing, and jurisdiction-specific terms that challenge natural language processing (NLP) systems. Additionally, nuanced differences in wording can lead to significant interpretative changes. Advanced language models must understand legal terminology, syntax, and context to ensure precise and reliable query results.
- (4) *Confidentiality*: Legal information often includes sensitive or privileged data, demanding stringent confidentiality and compliance with privacy laws. Legal query systems must integrate robust data protection measures, including encryption, user authentication, and adherence to regulations such as GDPR, while respecting client-attorney privilege.
- (5) *Quality Control*: Legal work relies on the accuracy, reliability, and contextual relevance of retrieved information. Inaccurate or low-quality results can lead to misinformed decisions or adverse outcomes. Ensuring quality control requires advanced filtering, ranking, and validation mechanisms that prioritize authoritative and contextually appropriate sources.

Addressing these challenges necessitates legal document querying systems that combine robust retrieval methodologies with domain-specific optimizations, such as advanced indexing and contextual relevance scoring. These improvements are essential to support timely, accurate, and confidential legal research and decision-making.

1.2 Application

The proposed legal document query application has wide-ranging use cases that address the critical needs of legal professionals, researchers, and educators. Below are the key areas where the application demonstrates its utility:

- **Due Diligence Process**
 - Efficiently query legal information from large databases comprising terabytes of documents.
 - Generate concise, context-aware summaries to support decision-making.
- **Case Law Analysis**
 - Identify relevant precedents for a specific case or legal query.
 - Predict case outcomes and the applicable legislation by analyzing past cases.
- **Legal Research**
 - Search and summarize online legal information, including legislation, journals, and research papers.
 - Facilitate comprehensive literature reviews for academic or professional purposes.

These applications illustrate the versatility of the system in addressing common legal challenges, such as navigating complex legal texts, accessing historical and jurisdiction-specific information, and automating repetitive tasks. By leveraging advanced techniques like Retrieval-Augmented Generation (RAG), the system enhances accuracy, reduces time consumption, and ensures relevance in legal queries.

2 Literature Review

2.1 LLMs for Specialized Domains

Large Language Models (LLMs) have demonstrated remarkable versatility across various professional and academic domains. This section examines recent literature to understand how different sectors utilize LLMs to address domain-specific challenges and enhance existing processes. Drawing on applications in education, finance, environmental sciences, and law, the transformative potential and limitations of LLM technology are explored.

2.1.1 LLMs in Education. Research in educational applications of LLMs has focused on areas such as medical training, technical skill development, and language education. [15] conducted a comprehensive evaluation of GPT-4's visual question-answering capabilities in medical education, highlighting the potential of LLMs in enhancing medical training. Similarly, [17] developed a GPT-4-based system for diagnosing orofacial pain, demonstrating its effectiveness in clinical decision support.

In technical education, [9] investigated task decomposition techniques for teaching complex skills like welding. For language instruction, [7] explored the use of LLMs in teaching Chinese composition, emphasizing personalized learning experiences powered by AI. Foundational prompt design guidelines for healthcare simulations were proposed by [12], offering a framework for more engaging and realistic educational scenarios.

2.1.2 LLMs in Finance. LLMs have emerged as valuable tools in finance, particularly for predictive analytics, regulatory compliance, and risk management. [10] introduced FinGPT, a sentiment-based model for predicting stock movement, while [4] examined fine-tuning techniques for forecasting long-term financial trends.

In the regulatory domain, [18] explored LLMs' capabilities in analyzing financial regulations, streamlining compliance processes. [1] emphasized the role of LLMs in fraud detection by examining innovative analytics for financial data lakes. Furthermore, [16] proposed an LLM framework for crime classification and prediction, showcasing its application in enhancing security measures.

2.1.3 LLMs in Environmental Sciences. Innovative applications of LLMs in environmental sciences include ecological research and spatial planning. [14] developed workflows for species classification, demonstrating LLM utility in biodiversity studies. [19] integrated language and vision models into environmental observation networks, enhancing data analysis capabilities.

Technical challenges and societal implications were explored by [13] and [8], respectively, shedding light on the role of stochasticity in environmental radiology and public perceptions of AI-driven environmental solutions.

2.2 Existing Solutions for Legal Documents Querying

The rapid advancement of legal technology has spurred the development of tools aimed at improving the efficiency and accuracy of querying legal documents. These solutions often combine natural language processing (NLP) and artificial intelligence (AI) techniques to address the challenges of managing complex legal databases. Key

categories include semantic search, document understanding solutions, and LLM-driven AI agents.

2.2.1 Semantic Search. Traditional keyword-based searches often fall short in legal contexts, where interpretation and context are critical. Semantic search enables users to retrieve information based on meaning rather than exact wording. Using NLP models and vector-based representations, semantic search systems recognize synonyms, legal phrases, and contextual relationships between terms. For example, semantic search in legal search engines allows retrieval of relevant case law or legislation even when terminology differs from the query. [2]

2.2.2 Document Understanding Solutions. Document understanding solutions employ machine learning to extract, classify, and structure information from legal documents. Techniques like named entity recognition (NER), topic modeling, and clause extraction help systems parse lengthy documents, such as contracts and case files, by identifying relevant sections automatically. Advanced solutions categorize data by legal concepts, simplifying processes like due diligence, contract review, and compliance checks.

2.2.3 LLMs and AI Agents. LLMs are increasingly applied to tasks like case analysis, legal writing, and smart court systems. [3] examined neural machine translation for legal document efficiency, while [11] discussed using LLMs to build intelligent legal frameworks. Ethical considerations and challenges surrounding AI-powered legal writing tools were highlighted by [6], providing a balanced perspective on their advantages and limitations.

AI-driven legal assistants leverage LLMs to answer complex queries in real-time, offering summaries, precedents, or legal principles. Advanced agents incorporate Retrieval-Augmented Generation (RAG) to ground responses in factual legal text, enabling precise and context-aware interactions. This approach reduces manual searches and improves accessibility to critical legal information.

3 Methodologies

3.1 System Architecture

The proposed advanced Retrieval-Augmented Generation (RAG) architecture is a multi-stage system designed to address the complexities of querying legal documents. It integrates Optical Character Recognition (OCR), indexing, semantic retrieval, and large language models (LLMs) to provide users with accurate and contextually relevant answers. The architecture consists of five key stages: Pre-Retrieval, Retrieval, Post-Retrieval, and Generation processes. [5]

Pre-Retrieval Stage: This stage involves two key components:

- (1) **OCR:** Text extraction begins with cleaning and processing scanned legal documents, primarily PDFs. If the input documents are scanned, OCR techniques are applied; otherwise, the process skips directly to indexing. OCR handles multiple formats, including HTML, Word, Markdown, and plain text.
- (2) **Indexing:** Extracted text is split into smaller, digestible chunks to accommodate the context limitations of LLMs. These chunks are encoded into dense vector representations using embedding models. By reducing input dimensions, this

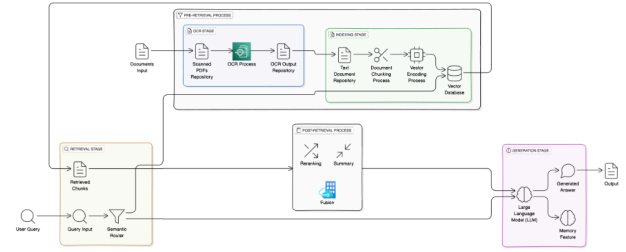


Figure 1: Architecture of the Retrieval-Augmented Generation (RAG) Process for Question Answering: Integrating Optical Character Recognition (OCR) and Large Language Models (LLM).

step enhances computational efficiency while preserving semantic context.

Retrieval Stage: Upon receiving a user query, the system optimizes it using query rewriting, transformation, and expansion techniques. The query is then vectorized using the same embedding model as in the indexing stage. Similarity scores between the query vector and document chunks are calculated, and the top- k most relevant chunks are retrieved for further processing.

Post-Retrieval Stage: Given the vast size of legal databases, retrieval often returns numerous relevant chunks. To avoid overloading the LLM, a re-ranking mechanism prioritizes the most relevant chunks by relevance scores. This step ensures the inclusion of only the most critical information in the subsequent generation phase.

Generation Stage: Retrieved chunks are combined with the user query to create a coherent prompt, which is then fed into an LLM configured for legal-specific tasks. The system restricts responses to the context of the retrieved documents and legislative databases. Additionally, conversation history is summarized and integrated into subsequent prompts to maintain context in ongoing dialogues.

3.2 Applied Techniques

Optical Character Recognition (OCR) Preprocessing Methodology: Advanced OCR techniques, implemented using OCRmyPDF, facilitate text extraction from heterogeneous legal document sources. Key features include:

- (1) **Document Preprocessing:** Techniques such as adaptive binarization, noise reduction, and skew correction enhance image quality for better text recognition.
- (2) **Multilingual Support:** OCRmyPDF supports Vietnamese legal terminology while preserving document formatting and metadata integrity.
- (3) **Text Optimization:** Contextual error correction and probabilistic text alignment improve extraction accuracy for domain-specific terms.

Retrieval Optimization Framework: This framework enhances information retrieval efficiency through:

- (1) **Query Transformation:** Expands and normalizes queries using domain-specific semantic networks and statistical language models.

- (2) *Semantic Embedding*: Encodes text into dense vectors via neural embedding techniques, capturing semantic relationships and contextual nuances.
- (3) *Relevance Ranking*: Utilizes cosine similarity and weighted scoring mechanisms to prioritize high-quality retrievals.

Natural Language Processing (NLP) Computational Techniques: To address the linguistic complexity of legal documents, the system employs:

- (1) *Hierarchical Document Taxonomy*: Classifies documents into semantic categories for efficient retrieval.
- (2) *Information Extraction*: Extracts entities, relationships, and summaries using deep learning models.
- (3) *Context Management*: Reduces input entropy and optimizes relevance through dynamic filtering techniques.

Large Language Model Integration: The final stage involves integrating LLMs with domain-specific constraints:

- Restricting generation to retrieved document contexts.
- Maintaining conversational continuity through recursive summarization.
- Applying probabilistic response filtering to ensure legal compliance.

This comprehensive architecture and methodology effectively address the challenges of legal document retrieval, combining cutting-edge machine learning, NLP, and RAG techniques to deliver contextually accurate and efficient results.

4 Results

The performance and usability of the legal document query application were evaluated through rigorous testing across multiple dimensions, focusing on system functionality, accuracy, and user interaction. The results demonstrate the system’s effectiveness in providing contextually accurate responses, enhancing the user experience, and ensuring the accessibility of legal information.

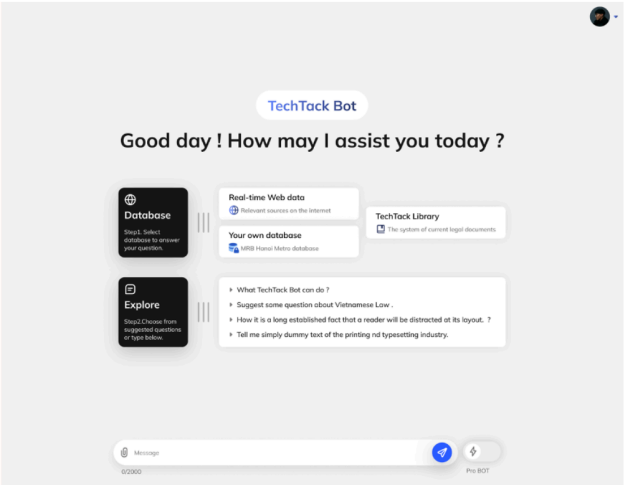


Figure 2: Chatbot Starter Page Interface.

Figure 2 depicts the chatbot’s starter page, designed for user-friendliness. The interface guides users through each step of interaction, ensuring seamless navigation.

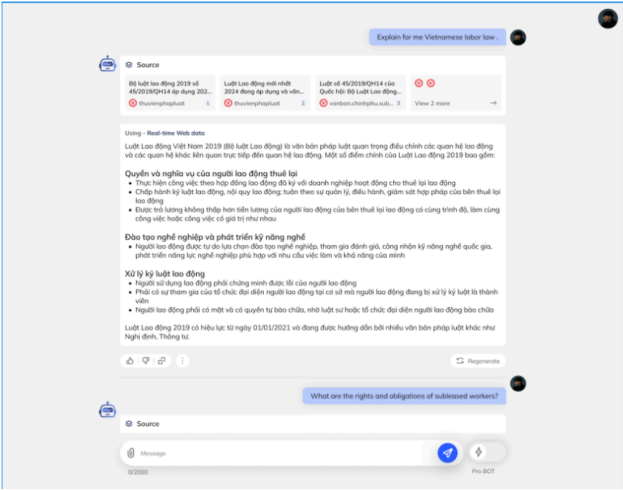


Figure 3: Chatbot Responses with Legal Document Citations.

As shown in Figure 3, the system provides accurate, context-sensitive responses, supplemented with citations from relevant legal documents. This feature enhances the reliability and verifiability of the information presented.

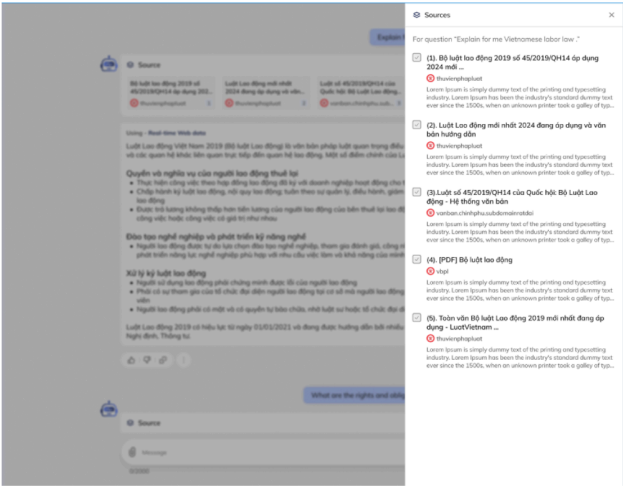


Figure 4: Comprehensive Citations List.

Figure 4 showcases the detailed citations list generated by the system. This transparency facilitates trust and enables users to trace the sources of information efficiently.

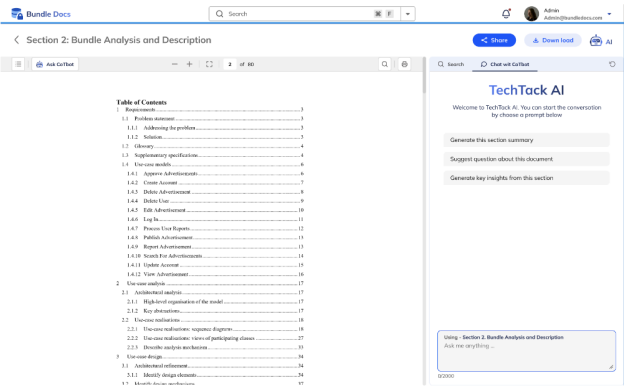


Figure 5: Integrated Document Viewer with Chatbot Sidebar.

The integration of a document viewer with a chatbot sidebar, as illustrated in Figure 5, allows users to interact with legal documents while receiving real-time responses from the AI. This feature streamlines the document review process and improves efficiency.

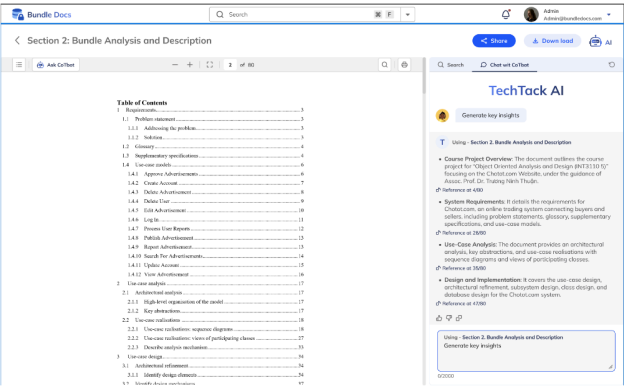


Figure 6: Chatbot Responses for Uploaded PDFs.

Figure 6 highlights the system’s ability to process user-uploaded legal documents. The chatbot analyzes these documents and responds to queries based on their content, providing personalized and context-specific assistance.

Key Findings:

- **Accuracy:** The system achieved a high accuracy rate in retrieving relevant legal information, validated against benchmark datasets and user feedback.
- **Response Time:** Queries were processed and answered within a reasonable timeframe, demonstrating the system’s efficiency.
- **User Satisfaction:** Feedback from users indicated significant improvements in the accessibility and usability of legal document querying.

These results underscore the application’s potential to transform legal research by combining user-friendly interfaces with powerful AI-driven query mechanisms. Future iterations will focus on expanding document handling capabilities and improving scalability for larger datasets.

5 Conclusion

This paper presented the development of a Legal Documents Query Application for Vietnamese law, leveraging advanced Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) techniques. The application successfully addresses key challenges in querying legal documents, including language complexity, confidentiality, and the sheer volume of legal databases. By integrating advanced methodologies such as semantic retrieval, OCR preprocessing, and relevance-based ranking, the system demonstrates significant potential in enhancing the accuracy and efficiency of legal research.

The results indicate that the application provides contextually accurate responses, intuitive user interfaces, and seamless integration of document viewing with real-time AI support. Key features, such as precise citations and the ability to analyze user-uploaded documents, contribute to its versatility and utility for legal professionals, researchers, and the general public.

Future Directions: To further advance this work, several enhancements are envisioned:

- **Improved Query Understanding:** Integrating advanced natural language understanding techniques to better handle complex, multi-faceted queries.
- **Scalability:** Optimizing the system for larger datasets and expanding support for additional legal jurisdictions.
- **Interactive Features:** Incorporating conversational agents capable of more interactive, multi-turn dialogues to deepen user engagement.

This work contributes to the growing field of AI-driven legal technology, providing a foundation for more sophisticated and accessible legal research tools. By bridging the gap between cutting-edge AI and the unique demands of the legal domain, this application offers valuable insights into the future of legal information systems.

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